All-Prime Borel j_! Upgrade

Autonomous discussion Pass 93 upgrades the finite-support Borel $j_!$ class to the honest all-prime $\operatorname{Spec}\mathbb{Z}$ setting.

The generic singleton $\{\eta\}$ is not open in honest $\operatorname{Spec}\mathbb{Z}$: every nonempty basic open $D(n)$ contains all but finitely many closed primes. Thus the finite-support $j_!$ notation must be read all-prime as a pro-open, continuous, or solid coefficient, not as ordinary extension by zero from an open generic point.

The all-prime coefficient is

$$\mathfrak{B}_{j!}^{\text{cont}} = \mathbb{Q}^\times \ltimes R\varprojlim_{S \Subset \mathbb{P}} j_{S,!}\mathcal{V}_S,$$

where S ranges over finite prime supports and $\mathcal{V}_S$ is the dilation coefficient from the finite recollement model.

For $S \subseteq T$, support restriction gives a surjection

$$\mathbb{Z}^T/\Delta\mathbb{Z} \to \mathbb{Z}^S/\Delta\mathbb{Z}$$

with kernel rank $|T| - |S|$; modulo N its kernel has size $N^{|T|-|S|}$. The support direction is therefore Mittag-Leffler and adds no new $\varprojlim^1$. The nonzero derived content remains the per-prime dilation tower inside $\mathcal{V}$.

The all-prime identity is:

$$H^1_{\text{cont}}(\operatorname{Spec}\mathbb{Z}, j_!\mathcal{V}) \cong \varprojlim_{S \Subset \mathbb{P}} H^1(X_S, j_{S,!}\mathcal{V}_S) \cong \widehat{\mathbb{Z}}/\mathbb{Z}.$$

The global Levi $\mathbb{Q}^\times$ is retained. Replacing it by a product of local Levi factors would be the local Loeb sheafification, not the Rosser/Borel torsor. Pushing out along $\mathbb{Z} \to \mathbb{Q}$ gives

$$0 \to \mathbb{Q} \to \mathbb{A}_f \to \widehat{\mathbb{Z}}/\mathbb{Z} \to 0,$$

and the hyperbolic Borel $\mathbb{Q}^\times \ltimes \epsilon$ acts on this same continuous shear class.

The finite checker `artifacts/reports/pass93-all-prime-borel-jshriek-upgrade-check.json` verifies the open-point distinction, support projection ranks, finite mod-N kernels, support-direction Mittag-Leffler behavior, and the all-prime Borel coefficient with unipotent limit $\widehat{\mathbb{Z}}/\mathbb{Z}$.